

# Complex Analysis Qualifying Exam Review

Fall 2026

August 15, 2026

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# Chapter 1

## Fall 2025

**Problem 1.1.** 5 Let  $\{f_n\}$  be a sequence of holomorphic functions on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that  $f_n \rightarrow f$  uniformly on compact subsets of  $\mathbb{D}$ , where  $f : \mathbb{D} \rightarrow \mathbb{C}$  is continuous. Prove that  $f$  is holomorphic on  $\mathbb{D}$ .

*Proof.* Morera's theorem. □

**Problem 1.2.** 6 Use contour integration to evaluate

$$\int_0^\infty \frac{x^2}{x^4 + 1} dx.$$

*Proof.* Use residue theorem for  $z_1 = e^{\frac{\pi i}{4}}, z_2 = e^{\frac{3\pi i}{4}}$ . □

**Problem 1.3.** 7 Let  $f$  be an analytic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \geq \sqrt{|z|}$$

for all  $z \in \mathbb{D}$ . Show that

$$|f(z)| \geq 1 \quad \text{for all } z \in \mathbb{D}.$$

**Problem 1.4.** 8 Determine how many solutions

$$z^7 - 7z^4 + z^3 - 3z = 1$$

are in the annulus

$$1 < |z| < 2.$$

*Proof.* Rouché's theorem. □

## Chapter 2

# Spring 2025

**Problem 2.1.** Let  $f(z)$  be holomorphic on  $\mathbb{C} \setminus \mathbb{R}$ .  $f$  is continuous at each point of  $\mathbb{R}$ . Show that  $f$  extends to an entire analytic function. (Hint: Use Morera's theorem.)

**Problem 2.2.** Evaluate

$$\text{p. v.} \int_{-\infty}^{\infty} \frac{e^{ix}}{x^2} dx$$

by residue theory.

**Problem 2.3.** Let  $f : \Omega \rightarrow \mathbb{C}$  be a non-constant holomorphic function defined on a connected open set  $\Omega \subset \mathbb{C}$ , and suppose that  $f$  is open (i.e., maps open sets to open sets). Additionally, suppose that for some  $z_0 \in \Omega$ , we have  $f'(z_0) = 0$ . Show that  $f$  is not injective on  $\Omega$ .

**Problem 2.4.** Let  $f : \mathbb{D} \rightarrow \mathbb{C}$  be a holomorphic function on the open unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

and suppose that

$$|f(z)| \leq M \quad \text{for all } z \in \mathbb{D},$$

for some constant  $M > 0$ . Prove the estimate that for all  $z \in \mathbb{D}$ ,

$$|f'(z)| \leq \frac{M}{(1 - |z|)^2}.$$

## Chapter 3

# Cauchy's integral formula

**Problem 3.1.** Evaluate the following integral:

$$\int_0^\infty \frac{1 - \cos x}{x^2} dx$$

**Problem 3.2.** Let  $U$  be a region. Suppose  $f$  is holomorphic in  $U$ , and a sequence  $\{z_n\} \subset U$  converges to some  $z \in U$ . If  $f(z_n) = 0$  for all  $n$ , show that  $f \equiv 0$ .

**Problem 3.3** (Spring 2018, Q4). Let  $f$  be an entire function, suppose that for each  $z_0$ , the power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

has at least one coefficient  $a_n = 0$ . Show that  $f$  must be a polynomial.

*Proof.* It suffices to show that for some  $k \in \mathbb{N}$ ,  $f^{(k)} \equiv 0$  because

$$a_n = \frac{1}{n!} f^{(n)}(z_0)$$

We first observe that

$$\bigcup_{n \in \mathbb{N}} \{z : f^{(n)} = 0\} = \mathbb{C}$$

Then there must exist some  $k \in \mathbb{N}$  such that

$$\{z : f^{(k)} = 0\} \text{ is uncountable}$$

Thus  $f^{(k)}$  has uncountably many zeros, which implies  $f \equiv 0$ .

**Lemma 3.1.** Let  $f$  be a nonzero holomorphic function, then  $f$  cannot have uncountably many zeros.

*Proof.* Let  $Z = \{z \in \mathbb{C} : f(z) = 0\}$ , we know that  $Z$  contains no limit point. We can tile the complex plane using unit squares and suppose  $Z$  is uncountable, the intersection of  $Z$  with one of the tiles needs to be uncountable. This is a contradiction because this intersection is compact and discrete, hence must be finite. □

□

**Problem 3.4 (Fall 2019, Q2).** Let  $f$  be an entire function such that

$$\sup_{|z|>R} |f(z)| \leq AR^k + B$$

for positive constants  $A, B$  and all  $R$  large. Then show  $f$  is a polynomial with degree at most  $k$ .

*Proof.* This problem is a direct application of Cauchy's inequality. Writing  $f(z)$  as a power series around 0, we have

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$a_n = \frac{1}{n!} f^{(n)}(0)$$

It suffices to show that for any  $m > k$ , we have

$$f^{(m)}(0) = 0$$

Using Cauchy's inequality, we get

$$\begin{aligned} |f^{(m)}(0)| &\leq \frac{m! \sup_{|z|=R} |f(z)|}{R^m} \\ &\leq \frac{m!(AR^k + B)}{R^m} \rightarrow 0 \end{aligned}$$

as  $R \rightarrow \infty$  because  $m > k$ . Hence we are done.  $\square$

**Problem 3.5 (Fall 2018, Q4).** Suppose  $f, g$  are both entire functions such that  $|f(z)| \leq |g(z)|$  for all  $z \in \mathbb{C}$ . Show that there exists  $c \in \mathbb{C}$  such that  $f = cg$ .

*Proof.* We will use Liouville's theorem. It suffices to show that  $h = f/g$  continuously extends to an entire function. Suppose  $g(z_0) = 0$  for some  $z_0$ , then we know  $f(z_0) = 0$ , hence writing

$$f(z) = (z - z_0)^n s(z), \quad g(z) = (z - z_0)^m t(z)$$

where  $s(z_0) \neq 0, t(z_0) \neq 0$ . We must have  $n \geq m$ , suppose if not, i.e.,  $m > n$ , then

$$|s(z)| \leq |z - z_0|^{m-n} |t(z)|$$

which is a contradiction by letting  $z = z_0$ . Thus  $z_0$  is a removable singularity for  $h = f/g$ . This implies  $h$  can be extended to an entire function, as desired.  $\square$

**Problem 3.6.**

## Chapter 4

# Residue formula

**Problem 4.1.** Compute

$$\int_0^\infty \frac{1}{1+x^2} dx$$

**Problem 4.2.** Assume that  $f : \mathbb{C} \rightarrow \mathbb{C}$  is an entire function, not identically equal to 0, and let

$$Z = \{z \in \mathbb{C} : f(z) = 0\}.$$

Prove that if  $Z$  is unbounded, then  $f$  has an essential singularity at  $\infty$ .